

Orange Carotenoid Protein as a Control Element in an Antenna System Based on a DNA Nanostructure

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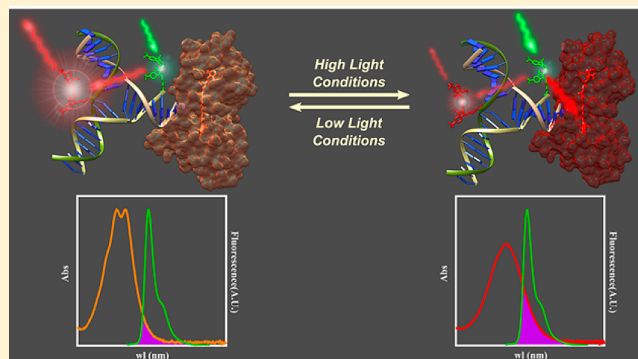
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S Supporting Information

ABSTRACT: Taking inspiration from photosynthetic mechanisms in natural systems, we introduced a light-sensitive photo protective quenching element to an artificial light-harvesting antenna model to control the flow of energy as a function of light intensity excitation. The orange carotenoid protein (OCP) is a nonphotochemical quencher in cyanobacteria: under high-light conditions, the protein undergoes a spectral shift, and by binding to the phycobilisome, it absorbs excess light and dissipates it as heat. By the use of DNA as a scaffold, an antenna system made of organic dyes (Cy3 and Cy5) was constructed, and OCP was assembled on it as a modulated quenching element. By controlling the illumination intensity, it is possible to switch the direction of excitation energy transfer from the donor Cy3 to either of two acceptors. Under low-light conditions, energy is transferred from Cy3 to Cy5, and under intense illumination, energy is partially transferred to OCP as well. These results demonstrate the feasibility of controlling the pathway of energy transfer using light intensity in an engineered light-harvesting system.

KEYWORDS: Orange carotenoid protein, DNA nanotechnology, artificial light-harvesting, photoprotection, fluorescence spectroscopy, energy transfer



Most attempts to mimic photosynthetic light-harvesting functions using engineered systems have focused on optimizing the energy-transfer efficiency, spectral coverage, and absorbance cross-section of the apparatus.^{1–6} In nature, however, an equally important aspect of light reactions is the extraordinary resilience to photodamage thanks to a series of mechanisms that intervene under light-stress conditions.⁷ Carotenoids are one of the key components in the photoprotective system of most photosynthetic organisms and have well-characterized structural, light-harvesting, and photoprotection roles.^{8–12} Within antennas, the organization of carotenoids around the chlorophyll molecules is critical for both energy transfer and photoprotection. Distance and orientation dictate the efficiency, and thus the time scale, of the energy transfer from an excited carotenoid to a chlorophyll, as well as the ability of the carotenoids to quench unwanted excited states of chlorophylls to prevent oxidative photodamage.¹³ Furthermore, under intense light illumination, many photosynthetic organisms including plants and green algae replace a fraction of their carotenoids (xanthophyll cycle),¹⁴ and in so doing switch between a light-harvesting mode and a photoprotection mode

depending on the light intensity. This is thought to happen through a conformational change induced by the replacement of the carotenoid, and experimental evidence for the direct deactivation of the excited chlorophyll by a carotenoid has been provided.^{14,15} In cyanobacteria, a soluble orange carotenoid binding protein (OCP) is employed as a light-sensitive regulatory element with a photoprotection role.^{16,17} The OCP is constitutively expressed but is only activated under high-light conditions, resulting in controlled energy flow from the phycobilisome, the antenna system in cyanobacteria, to the reaction centers.¹⁸ As recently reported, OCP is one of possibly multiple mechanisms that regulate the energy flow in the phycobilisome on different time scales.¹⁹ The protein consists of two domains, the N-terminal and the C-terminal domains, and a carotenoid molecule, 3'-hydroxyechinenone, is bound between them with limited exposure to the solvent.²⁰ The two domains are connected by a flexible linker loop, and an α -

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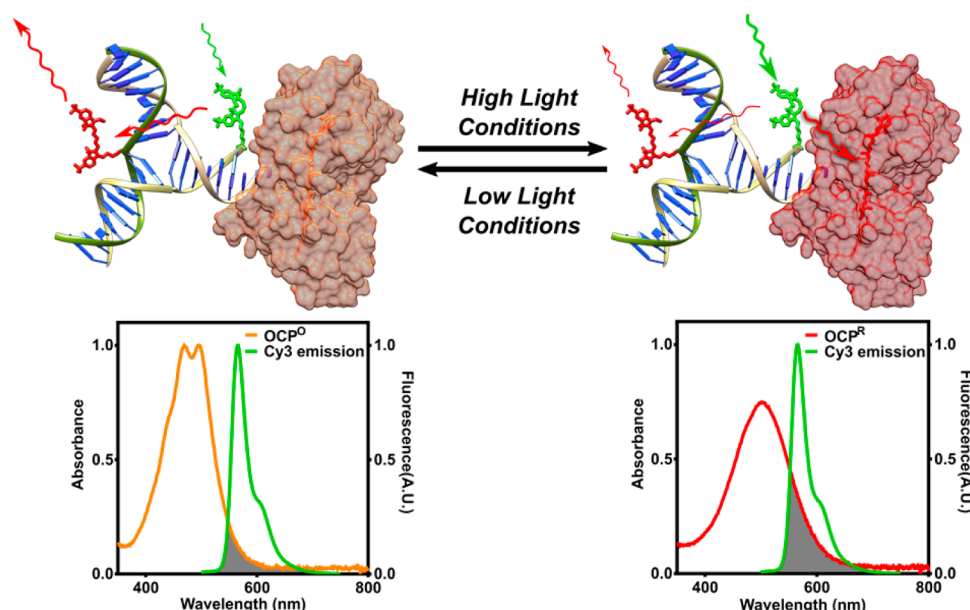


Figure 1. Top panel: schematics depicting the conceptual design of the OCP–DNA–dyes system used in this work. The arrows indicate the energy flow between the components. On the left, the system is represented under low-light illumination, and on the right, it is represented under high-light conditions. Bottom panel: the two panels show the spectra overlap between the emission of Cy3 and the absorption of OCP^O (left), and OCP^R (right). The spectral overlap between the Cy3 emission and OCP absorption is shaded in gray for clarity.

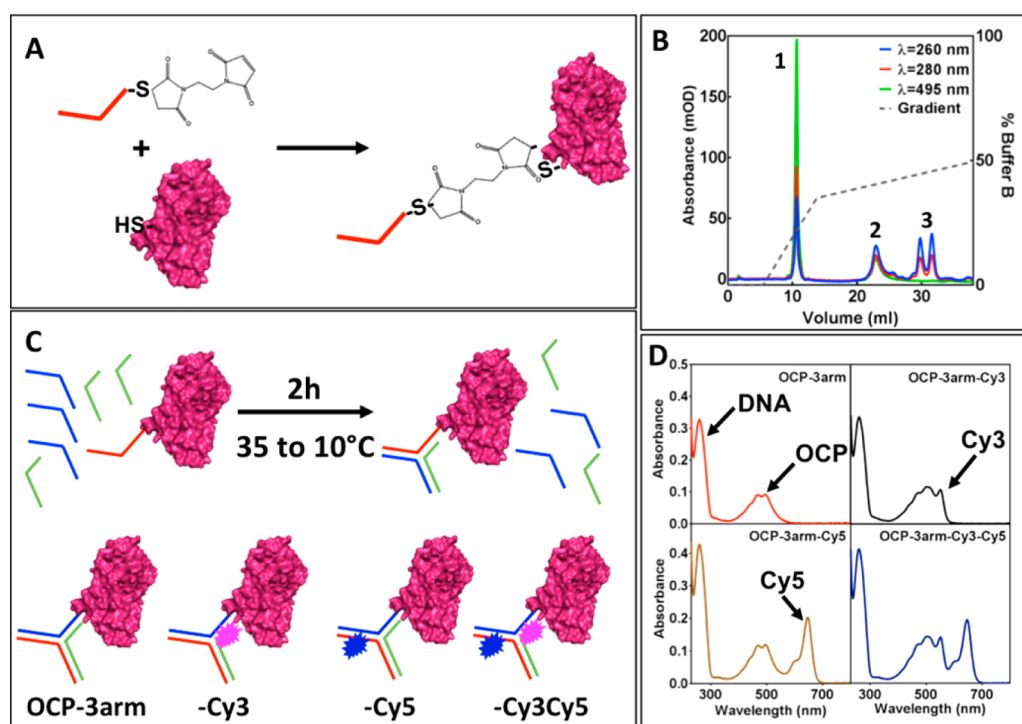


Figure 2. (A) Scheme of the coupling reaction used to link the DNA strand St1 with OCP. The maleimide-modified strand is reacted with the protein to form a stable, covalently attached, protein–DNA bioconjugate. (B) Anion-exchange chromatography results for the purification of OCP–St1 after the conjugation reaction. Absorbance measured at 260, 280 and 495 nm show the presence of DNA (blue), protein (red), and protein–DNA conjugate (green) in the eluate, respectively. The dashed line indicates the elution gradient. (C) Upper panel: scheme illustrating the annealing of the DNA three-arm junction on the OCP–St1 construct. Lower panel: the four OCP–3arm constructs that were separately produced by substituting the strands St2 and St3 with their labeled counterpart. (D) Absorption spectra of the four OCP–3arm samples after purification. The spectral signatures of DNA (three-arm), OCP, Cy3, and Cy5 were used to calculate the ratio between the components, as indicated in the main text.

helical arm (αA) extends from the N-terminal to the C-terminal domain that contributes to the control of the activity of the protein.²¹ Upon illumination at high light intensity, the protein switches from its orange, OCP^O, form to its red, OCP^R, form,

which involves a red shift of the absorption maximum of the carotenoid and the disappearance of its vibronic signature.¹⁷ Recent studies have elucidated the detailed structure and molecular mechanism that takes place during light activation of

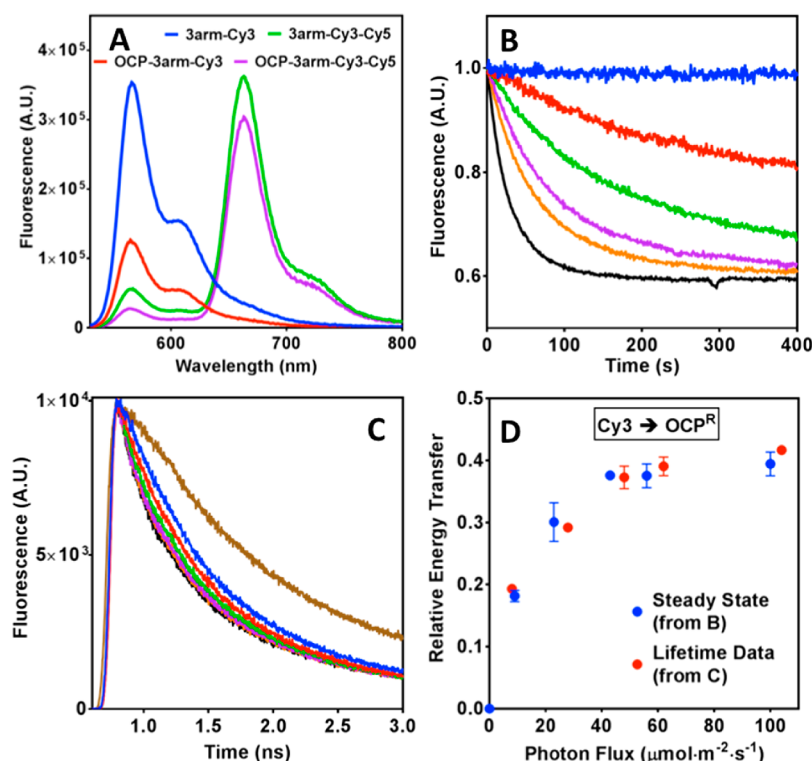


Figure 3. (A) Fluorescence emission spectra of samples used to calculate the energy-transfer efficiency from Cy3 to Cy5 and OCP. To calculate the efficiency between different components within a complex, the emission spectra were recorded at low illumination intensity with addition of FRP to maintain OCP in the orange state. (B) The fluorescence intensity change of the OCP-3arm-Cy3 sample as a function of exposure time monitored at 560 nm (Cy3 emission). The OCP-3arm-Cy3 was excited at 520 nm with photon fluxes of 9 (blue), 9 (red), 23 (green), 43 (magenta), 56 (orange), and 100 (black) $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, and the curves were normalized at the maximum intensity at time zero. The blue trace was recorded in the presence of FRP. (C) Emission decay curves at 560 nm showing the effect of light activation of OCP on the lifetime of the Cy3 dye. The brown curve is the decay kinetics of Cy3 in the 3arm-Cy3 sample. The blue, red, green, magenta, orange, and black traces were recorded with excitation densities of 8, 8, 28, 48, 62, 104 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, respectively. The blue trace was measured in the presence of FRP. (D) Relative emission of Cy3 was calculated from the data in panels B (blue circles) and C (red circles) as a function of excitation intensity. The OCP+FRP sample was used as a reference to calculate the data point at a photon flux of 0 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. Excitation was at 520 nm for all measurements.

the OCP as well as the associated conformational changes it undergoes throughout the process. Upon light excitation, the carotenoid molecular conformation apparently changes, and its interaction with the surrounding protein residues is perturbed.^{17,22} This leads to a movement of the αA arm away from the C-terminal domain; subsequently the C- and N-terminal domains dissociate and the carotenoid translocates into the N-terminal domain.^{21,23–26} The translocation of the carotenoid partially exposes its extremities to the solvent and modifies the environment around it, presumably causing the change in the absorption features observed during photoconversion from OCP^O to OCP^R.^{27,28} Energetically, both the red and orange form of OCP are able to quench the fluorescence of the phycobilisome antenna.^{29,30} However, the distance and orientation of OCP relative to the phycobilisome dictates the quenching efficiency. Past work has demonstrated that the protein conformational changes that occur upon light activation enable the association between OCP^R and the phycobilisome but not OCP^O.^{31,32} This docking event is thought to bring the carotenoid close enough to the pigments in the antenna complex to quench their fluorescence, although the precise mechanism of quenching has yet to be established.³³

Here, we will explore the functional incorporation of the OCP into an artificial antenna system. The programmability, biocompatibility, and modularity of DNA make it an ideal building material for the construction of complex networks of

photonic elements. DNA nanostructures and DNA origami structures have been successfully used to build photonic systems that mimic photosynthetic function, and to assess their viability as a starting point for larger photonic assemblies.³⁴ Several examples of biohybrid antennas have recently been presented (see refs 35 and 36 for an overview). In the current work, a three-strand DNA junction³⁷ (3arm) is used for the modular assembly of a simple antenna structure that incorporates OCP and exploits its light-responsive features to control excitation energy flow (Figure 1, top). This work focuses on using the spectral shift of the OCP cofactor upon photoconversion as a means of introducing a light-dependent modulation of energy-transfer efficiency between two target dyes. Although the detailed mechanism of modulation is somewhat different from the natural function of OCP (in which it is predominantly the movement of the cofactor that modulates energy transfer), this provides a conceptual demonstration of principle. The dye pair Cy3–Cy5 is used in this study as an energy donor and acceptor, respectively. There is a substantial increase in spectral overlap between emission of the Cy3 dye and the absorption of OCP when the OCP is in the red form (Figure 1, bottom), making the OCP^R a reasonable energy-transfer acceptor from Cy3. Excitation in the 500–560 nm region will both excite Cy3 and activate the OCP. As the light-intensity level increases, the amount of OCP^R will increase and therefore quench Cy3 excited states

more effectively (the carotenoid in OCP has a very short excited state lifetime³⁰). Thus, the excitation of Cy3 (and OCP) at low power should result in an efficient Cy3-to-Cy5 energy transfer (OCP remains largely inactive at low intensity), but high-intensity excitation should activate a larger fraction of the OCP resulting in rapid energy transfer between the Cy3 excited state and OCP^R, providing a pathway that competes with energy transfer from Cy3 to Cy5, in analogy to natural photosynthetic photoprotection mechanisms.

The conjugation of the DNA strand St1 to OCP was performed using maleimide chemistry (Figure 2A). The final conjugate was separated from the unreacted components using anion-exchange chromatography (Figure 2B), and the OCP–St1 bioconjugate was annealed with two other strands to form the 3arm DNA structure. The use of a three-way junction provides a modular system, and four combinations of photonic elements were prepared by choosing the appropriate combination of modified DNA strands (Figure 2C): OCP with no dye (OCP–3arm), OCP with Cy3 only, OCP with Cy5 only, and OCP with both Cy3 and Cy5. Analogous 3arm structures without OCP were used as controls (see the Supporting Information for details). The final product, purified by means of size-exclusion chromatography (Figure S1), shows distinct spectral signatures of the DNA, OCP, Cy3, and Cy5 (Figure 2D). After spectral deconvolution, an analysis of the absorbance at the corresponding wavelength of each component indicates that their relative stoichiometry is about 1:0.8:0.8:0.9 = OCP/DNA/Cy3/Cy5 (see the Supporting Information for extinction coefficients; see Figure S3 for more details), and gel electrophoresis in native conditions confirm the co-assembly of DNA, labels, and protein in the OCP–DNA–dyes constructs (Figure S2). The photoconversion activity of the DNA conjugated OCP was tested and compared with unconjugated OCP under the same conditions using 380–480 nm illumination (430 ± 50 nm bandpass filter; Figure S4). The OCP–3arm system retains the photoactive behavior of the unmodified OCP, i.e., it can be converted to its red form (OCP^R–3arm), and it converts back to the orange form, (OCP^O–3arm) albeit with slower kinetics than those of the unconjugated protein.

Energy-transfer efficiencies from Cy3 to Cy5 and from Cy3 to OCP in the OCP–3arm constructs were determined by keeping OCP in its orange form (OCP^O) using the fluorescence recovery protein (FRP)^{38,39} (see Figure S5 for further details). The steady-state fluorescence emission spectra of OCP^O–3arm–Cy3 and OCP^O–3arm–Cy3–Cy5 were recorded under these conditions and compared with the reference 3arm–dye samples that do not contain OCP in Figure 3A. Upon excitation at 520 nm and using the 3arm–Cy3 sample as a reference, the relative fluorescence emission of each of the other samples was calculated and is reported in Table 1. A list of the corresponding energy-transfer values is shown in Table S2. In the presence of OCP^O or Cy5, the emission of Cy3

is reduced to 0.4 and 0.15 times its original value, respectively, in comparison to the fluorescence from 3arm–Cy3 alone. Based on this, one would expect that when both OCP^O and Cy5 are present (i.e., OCP^O–3arm–Cy3–Cy5), the emission of Cy3 should reduce to a relative value of 0.11 (i.e., $1/(1/0.15 + 1/0.4)$) because both energy-transfer paths are working simultaneously. The experimental data show reduction to a relative value of 0.08, which is in reasonable agreement with that expected. These results indicate that OCP^O and Cy5 both behave as independent acceptors of the energy from Cy3, and when both are present (OCP^O–3arm–Cy3–Cy5 sample) the overall Cy3 → [Cy5, OCP] energy transfer is approximately the sum of the transfer rates of the single components.

Constant illumination of OCP–3arm–Cy3 at 520 nm excites Cy3, resulting in the steady-state photoconversion of OCP to the red (OCP^R) form. OCP^R quenches the fluorescence of Cy3 more efficiently than does OCP^O due to an increased spectral overlap between Cy3 emission and OCP^R absorbance, relative to Cy3 and OCP^O (Figure 1).

In Figure 3B, the fluorescence at 566 nm for the OCP–3arm–Cy3 sample is monitored under continuous illumination using different intensities of light. As a control, the result of including FRP (4-fold excess over OCP) to maintain the OCP in its orange form is also shown (blue line). In the absence of FRP, the fluorescence of Cy3 in this sample decreases during illumination until it reaches a plateau. As the light intensity increases, the rate of decrease is faster, and the final steady-state fluorescence level is lower. The sample with FRP, which should hold OCP in the orange form, does not show a time-dependent decrease. As another control, the fluorescence from Cy3 in the 3arm–Cy3 construct without OCP was also measured and is presented in Figure S11; as expected, this sample showed no decrease in fluorescence intensity with time over the range of intensities used. The results of Figure 3B are in line with the concept that under more-intense illumination, the conversion from OCP^O to OCP^R takes place more rapidly, and the final steady-state population of OCP^R becomes larger. The data suggests that the more OCP^R present, the more Cy3 fluorescence is quenched. The end points of the time traces in Figure 3B were used to calculate the relative amount of energy transfer occurring via the reaction Cy3 → OCP^R at different illumination intensities, using the emission of the OCP^O–3arm–Cy3 sample as a reference, and the results are plotted in Figure 3D as a function of photon flux (blue symbols). In analogy with energy-transfer efficiency, the relative energy transfer here is defined as $E = 1 - I_{\text{Cy3, OCP}^{\text{R}}}/I_{\text{Cy3, OCP}^{\text{O}}}$, where the $I_{\text{Cy3, OCP}^{\text{O}}}$ is the fluorescence of Cy3 when OCP is kept in its orange form, and $I_{\text{Cy3, OCP}^{\text{R}}}$ is the fluorescence of Cy3 at the end point of the time traces presented in Figure 3B.

The time-resolved fluorescence decay of the Cy3 dye was measured using time correlated single photon counting techniques. Similar light intensities were used in these measurements as those used in the constant illumination experiments (Figure 3B). The Cy3 fluorescence decay traces as a function of excitation intensity are shown in Figure 3C. The fluorescence decay kinetics of 3arm–Cy3 is also shown as a reference (brown line). The partial spectral overlap between OCP^O and Cy3 results in a reduced Cy3 lifetime in the OCP^O–3arm–Cy3 sample compared to that of 3arm–Cy3. Therefore, OCP^O–3arm–Cy3 (blue line) was used as the reference for calculation of the relative energy-transfer efficiency in each sample (Figure 3D, red symbols). The lifetime of OCP^R–3arm–Cy3 (in the absence of FRP) shortens as a function of

Table 1. Relative Fluorescence Emission Intensity at 662 nm of Samples Calculated Using 3arm–Cy3 as a Reference

sample	relative fluorescence
3arm–Cy3	1
OCP ^O –3arm–Cy3	0.4
3arm–Cy3–Cy5	0.15
OCP ^O –3arm–Cy3–Cy5	0.08

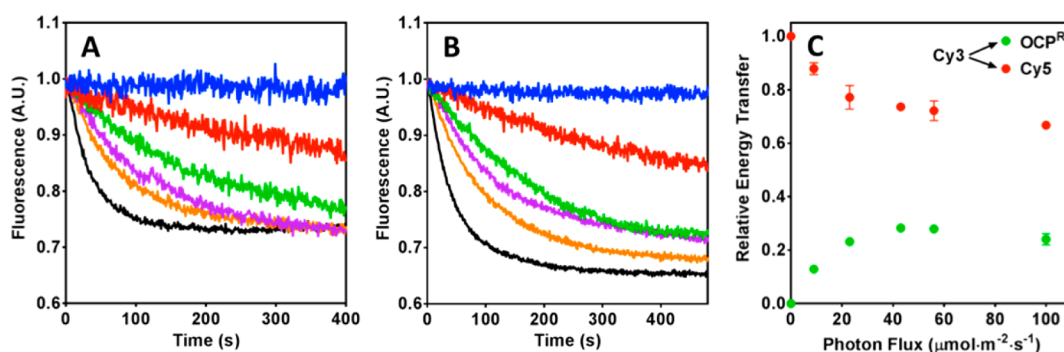


Figure 4. (A) Fluorescence intensity of Cy3 in the OCP–3arm–Cy3–Cy5 sample recorded at 560 nm upon excitation at 520 nm with photon fluxes of 9 (blue), 9 (red), 23 (green), 43 (magenta), 56 (orange), and 100 (black) $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. The blue curve was recorded in the presence of an excess of FRP. (B) Fluorescence intensity of Cy5 at 662 nm in OCP–3arm–Cy3–Cy5 upon excitation at 520 nm at various intensity levels. The color coding is the same as for panel A. (C) Energy transfer from Cy3 to OCP (green) and from Cy3 to Cy5 (red) extracted from the data in panels A and B, reported as a function of illumination intensity.

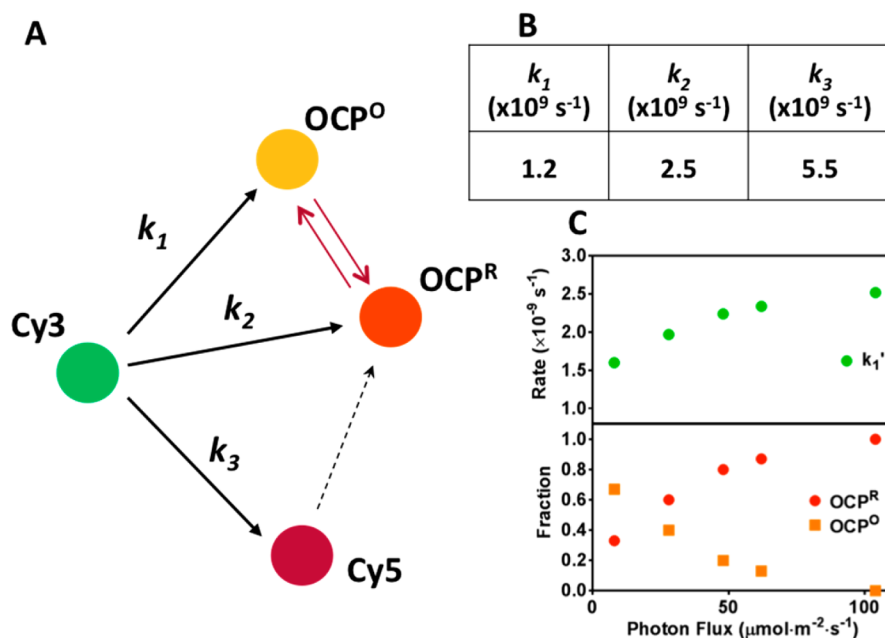


Figure 5. (A) A kinetic model for the energy-transfer pathways in the OCP–3arm–Cy3–Cy5 system. The Cy3 dye is the main donor when the 520 nm excitation light is used, and OCP^O, OCP^R, and Cy5 are the possible acceptors. The black arrows represent the energy transfer paths with the associated rate constants. The red arrows represent the interconversion between the two forms of the protein. The dashed arrow from Cy5 to OCP^R represents the negligible transfer from the Cy5 dye to OCP^R. (B) Rate constants were extracted from the TCSPC data. (C) Upper panel: the apparent rate constant k_1' as a function of excitation intensity, calculated from the data in Figure 2C. Lower panel: the relative populations of OCP in the orange (OCP^O, solid square) and red (OCP^R, solid circle) states.

increasing excitation intensity. The amplitude-weighted average lifetimes obtained from data fitting with a three-exponential decay model are presented in Table S5 and were used to calculate the relative energy transfer in the samples as a function of excitation intensity, with $E = 1 - \tau_{\text{Cy3, OCP}^R} / \tau_{\text{Cy3, OCP}^O}$, where $\tau_{\text{Cy3, OCP}^O}$ is the lifetime of Cy3 when OCP is in the orange form, and $\tau_{\text{Cy3, OCP}^R}$ is the lifetime of Cy3 when OCP is (partially) converted to the red form (Figure 3C and Table S5). This result was then compared to the efficiencies calculated using the continuous illumination data in Figure 3D. Both measurements, steady-state emission and lifetime data, show very similar results, indicating that the energy-transfer efficiency from Cy3 to OCP^R increases as a function of illumination intensity due to the accumulation of OCP^R. In particular, both the fluorescence intensity and its average lifetime decrease with increasing illumination intensity. As OCP is converted from the

orange to the red form via illumination, it becomes better at accepting energy from Cy3, thus quenching the emission of the dye. The response of OCP to light activation shown here is proportional to the amount of light reaching the sample in a nonsaturating range, as previously reported from absorption measurements.¹⁷ It can be seen from Figure 3D that at a photon flux of $\sim 100 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, the energy-transfer efficiency from Cy3 to OCP^R levels off at a value of ~ 0.4 .

The ability of OCP to quench the excited state of the donor in competition with energy transfer to an acceptor was investigated further in the OCP–3arm–Cy3–Cy5 system. As shown in Figure 3B, illumination at 520 nm induces photoconversion of OCP^O to OCP^R and quenches the emission from Cy3 in the OCP–3arm–Cy3 construct. In the OCP–3arm–Cy3–Cy5 construct, emission from Cy3 is light-dependent as well (Figure 4A), at least partially due to

quenching by OCP^R as observed in the OCP–3arm–Cy3 system. An intensity decrease in the emission of Cy5 in the OCP–3arm–Cy3–Cy5 construct was also observed as a function of increasing light intensity (Figure 4B). In this case, there are two energy-transfer pathways from Cy3, i.e., to OCP and to Cy5. The relative efficiency of the Cy3 → OCP energy transfer was calculated as $E_{\text{Cy3} \rightarrow \text{OCP}^{\text{R}}} = 1 - I_{\text{Cy3, OCP}^{\text{R}}} / I_{\text{Cy3, OCP}^{\text{O}}}$, where $I_{\text{Cy3, OCP}^{\text{O}}}$ is the fluorescence intensity of Cy3 in the presence of the orange form of OCP, and $I_{\text{Cy3, OCP}^{\text{R}}}$ is the emission of Cy3 when OCP is activated to its red form at different light intensities. The relative energy-transfer efficiency for the Cy3 → Cy5 pathway can be calculated using the emission of the acceptor as an indication of the energy received from Cy3, and thus, $E_{\text{Cy3} \rightarrow \text{Cy5}} = I_{\text{Cy5, OCP}^{\text{R}}} / I_{\text{Cy5, OCP}^{\text{O}}}$, where $I_{\text{Cy5, OCP}^{\text{O}}}$, $I_{\text{Cy5, OCP}^{\text{R}}}$ are the emission intensities of Cy5 in the presence of the orange form and the red form of OCP, respectively. The energy transfer from Cy3 to the photo-activated OCP as a function of illumination intensity follows the same trend observed for OCP–3arm–Cy3 construct under the same experimental conditions (Figure 3D). The decrease in fluorescence emission of Cy5 that is observed in Figure 4B is once again likely due to a depletion of energy from Cy3 by OCP^R and not by a direct energy transfer from Cy5 to OCP^R (the spectral overlap being negligible, Figure S7). Also, the energy-transfer efficiency for Cy3 → OCP is less in OCP–3arm–Cy3–Cy5 than that in OCP–3arm–Cy3 (~0.3 versus ~0.4). The lifetime data for Cy3 in the OCP–3arm–Cy3–Cy5 are more complex than for OCP–3arm–Cy3. The average lifetime is indeed shorter compared to the 3arm–Cy3–Cy5 sample (Figure S6A), but only the shorter lifetime components ($\tau < 0.5$ ns) seem to be substantially affected by the energy transfer from Cy3 to OCP^R, decreasing in lifetime with increasing illumination intensity as one would expect (Figure S6C).

The energy-transfer efficiencies (Table S2) obtained from fluorescence-decay measurements were used to calculate the energy-transfer rate constants in the OCP–3arm–Cy3–Cy5 system. Figure 5A presents a model for the energy-transfer pathways between the donor and the acceptors, with each transfer represented by a black arrow and the interconversion of OCP in its two forms represented by two red equilibrium arrows. The dashed arrow from Cy5 to OCP^R indicates that the transfer between the dye and the active form of the protein (further details in Figure S7) appears to be small enough not to affect the final results of the calculations. The rate constants k_1 , k_2 , and k_3 represent the microscopic energy-transfer rate constants in the system, and they can be calculated from the lifetime and energy transfer efficiency data. Their values are reported in Figure 5B (see the Supporting Information for details on the calculations). Modeling the data in this way indicates that the energy-transfer rate from Cy3 to OCP increases about 2.2-fold upon conversion of the protein from the inactive (OCP^O) to the active (OCP^R) state. The ~2-fold increase of k_2 ($2.5 \times 10^9 \text{ s}^{-1}$) compared to k_1 ($1.2 \times 10^9 \text{ s}^{-1}$) takes it closer to the Cy3 → Cy5 transfer rate, k_3 ($5.5 \times 10^9 \text{ s}^{-1}$) (Figure 5B), and allows energy transfer to the OCP's carotenoid to be more effective in OCP^R than in OCP^O, though not as effective as to Cy5. The results shown in Figure 4B, where a decrease of Cy5 emission is seen to behave in the same way as for Cy3, illustrate that OCP^R competes with Cy5 as an energy-transfer acceptor more effectively than it does in the OCP^O form. For the OCP–3arm–Cy3 sample, the Cy3 fluorescence lifetime data provide an average lifetime that

results from a contribution of k_1 , k_2 , and the relative populations of OCP in either the OCP^O or OCP^R states. Figure 5C shows the apparent rate constant k'_1 ($= 1/\tau_{\text{avg}}$) as extracted from the fluorescence lifetime measurements. We can calculate the fraction of OCP in the OCP^O and OCP^R forms by using the equation: $k'_1 = (1 - \alpha) k_1 + \alpha k_2$, where α is the fraction of OCP^R and $(1 - \alpha)$ represent the fraction of OCP^O. The data were used to produce the plot in the lower panel of Figure 5C that shows the composition of the sample as a function of illumination intensity.

In summary, this study has demonstrated that the orange carotenoid protein from cyanobacteria can be incorporated as a light-regulating component into an artificial antenna system using DNA as a scaffold. When conjugated to DNA, OCP retains its capability to switch back and forth from the orange to the red form in response to light activation, and it does so in an intensity dependent manner that is subject to FRP regulation. The data presented show that the protein is able to quench the fluorescence of a proximal dye donor, Cy3, and as a consequence competes for energy transfer to Cy5. Therefore, OCP can be used as a quenching element in an artificial antenna to regulate energy-transfer pathways in response to light-intensity changes. It is possible to foresee the incorporation of the protein into more-complex energy-transfer networks, organized for example on larger DNA origami structures. In a network of light-harvesting elements, OCP could provide photoprotection by limiting the excited-state lifetime at high-intensity illumination while preserving efficient energy capture at low to moderate illumination intensity. The short lifetime of the carotenoid (a few picoseconds) should allow OCP to efficiently quench several fluorophores in its proximity, therefore limiting the number required in the system; the complexity of designs can thus be reduced. Designs resembling the natural system, in which a tighter control of distances may lead to high quenching efficiency of a larger number of fluorophores, are in principle possible, although a more-refined origami structure would be required to accomplish this. Ultimately, the integration of other photochemically responsive elements, such as reaction-center proteins that perform charge separation upon light activation, would open the possibility of designing artificial photosystems for specific purposes and with a photoprotection mechanism that functionally mimics the natural one.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.6b04846.

Additional details on materials and methods, data-analysis procedures, supplementary results and discussion, chromatography results, additional spectroscopic characterization of samples and controls with steady-state and time-correlated single photon counting methods, gel electrophoresis results, appendix calculations, and supplementary data tables and figures. (PDF)

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Author Contributions

A.A., S.L., and N.W.W. conceived the research project. A.A. and S.L. designed the experiments with inputs from the other authors. H.L. and R.E.B. prepared the OCP and FRP samples and provided instructions on how to handle and use the proteins. A.A. prepared the OCP samples modified with DNA and performed the spectroscopic measurements on OCP-DNA and DNA samples. A.A. and S.L. analyzed the data. All the authors contributed in the discussion of the data. A.A. and S.L. wrote the manuscript with input from all authors.

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Notes

The authors declare no competing financial interest.

■ ABBREVIATIONS

OCP, orange carotenoid protein; FRP, fluorescence recovery protein

■ REFERENCES

- Balzani, V.; Credi, A.; Venturi, M. *Curr. Opin. Chem. Biol.* **1997**, *1* (4), 506–513.
- Wasielewski, M. R. *Acc. Chem. Res.* **2009**, *42* (12), 1910–1921.
- Marek, P. L.; Hahn, H.; Balaban, T. S. *Energy Environ. Sci.* **2011**, *4* (7), 2366–2378.
- Frischmann, P. D.; Mahata, K.; Wurthner, F. *Chem. Soc. Rev.* **2013**, *42* (4), 1847–1870.
- Gust, D.; Moore, T. A.; Moore, A. L. *Faraday Discuss.* **2012**, *155*, 9–26.
- Harriman, A. *Chem. Commun.* **2015**, *51* (59), 11745–11756.
- Blankenship, R. E. *Molecular Mechanisms of Photosynthesis*; Blackwell Science Ltd: Hoboken, NJ, 2002.
- Domonkos, I.; Kis, M.; Gombos, Z.; Ughy, B. *Prog. Lipid Res.* **2013**, *52* (4), 539–561.
- Kirilovsky, D. *Nat. Chem. Biol.* **2015**, *11* (4), 242–243.
- Scholes, G. D.; Fleming, G. R.; Olaya-Castro, A.; van Grondelle, R. *Nat. Chem.* **2011**, *3* (10), 763–774.
- Polivka, T.; Frank, H. A. *Acc. Chem. Res.* **2010**, *43* (8), 1125–1134.
- Demmig-Adams, B.; Adams, W. W. *Planta* **1996**, *198* (3), 460–470.
- Liu, Z.; Yan, H.; Wang, K.; Kuang, T.; Zhang, J.; Gui, L.; An, X.; Chang, W. *Nature* **2004**, *428* (6980), 287–292.
- Muller, P.; Li, X. P.; Niyogi, K. K. *Plant Physiol.* **2001**, *125* (4), 1558–1566.
- Ruban, A. V.; Berera, R.; Iliaia, C.; van Stokkum, I. H. M.; Kennis, J. T. M.; Pascal, A. A.; van Amerongen, H.; Robert, B.; Horton, P.; van Grondelle, R. *Nature* **2007**, *450* (7169), 575–U22.
- Wilson, A.; Ajlani, G.; Verbavatz, J. M.; Vass, I.; Kerfeld, C. A.; Kirilovsky, D. *Plant Cell* **2006**, *18* (4), 992–1007.
- Wilson, A.; Punginelli, C.; Gall, A.; Bonetti, C.; Alexandre, M.; Routaboul, J. M.; Kerfeld, C. A.; van Grondelle, R.; Robert, B.; Kennis, J. T. M.; Kirilovsky, D. *Proc. Natl. Acad. Sci. U. S. A.* **2008**, *105* (33), 12075–12080.
- Kirilovsky, D. *Photosynth. Res.* **2015**, *126* (1), 3–17.
- Gwizdala, M.; Berera, R.; Kirilovsky, D.; van Grondelle, R.; Krüger, T. P. *J. Am. Chem. Soc.* **2016**, *138* (36), 11616–11622.
- Kerfeld, C. A.; Sawaya, M. R.; Brahmandam, V.; Cascio, D.; Ho, K. K.; Trevithick-Sutton, C. C.; Krogmann, D. W.; Yeates, T. O. *Structure* **2003**, *11* (1), 55–65.
- Thurotte, A.; Lopez-Igual, R.; Wilson, A.; Comolet, L.; Bourcier de Carbon, C.; Xiao, F. G.; Kirilovsky, D. *Plant Physiol.* **2015**, *169* (1), 737–747.
- Wilson, A.; Punginelli, C.; Couturier, M.; Perreau, F.; Kirilovsky, D. *Biochim. Biophys. Acta, Bioenerg.* **2011**, *1807* (3), 293–301.
- Leverenz, R. L.; Sutter, M.; Wilson, A.; Gupta, S.; Thurotte, A.; Bourcier de Carbon, C.; Petzold, C. J.; Ralston, C.; Perreau, F.; Kirilovsky, D.; Kerfeld, C. A. *Science* **2015**, *348* (6242), 1463–1466.
- Liu, H. J.; Zhang, H.; King, J. D.; Wolf, N. R.; Prado, M.; Gross, M. L.; Blankenship, R. E. *Biochim. Biophys. Acta, Bioenerg.* **2014**, *1837* (12), 1955–1963.
- Gupta, S.; Guttman, M.; Leverenz, R. L.; Zhumadilova, K.; Pawlowski, E. G.; Petzold, C. J.; Lee, K. K.; Ralston, C. Y.; Kerfeld, C. A. *Proc. Natl. Acad. Sci. U. S. A.* **2015**, *112* (41), E5567–E5574.
- Liu, H. J.; Zhang, H.; Orf, G. S.; Lu, Y.; Jiang, J.; King, J. D.; Wolf, N. R.; Gross, M. L.; Blankenship, R. E. *Biochemistry* **2016**, *55* (7), 1003–1009.
- De Re, E.; Schlau-Cohen, G. S.; Leverenz, R. L.; Huxter, V. M.; Oliver, T. A. A.; Mathies, R. A.; Fleming, G. R. *J. Phys. Chem. B* **2014**, *118* (20), 5382–5389.
- Kish, E.; Pinto, M. M. M.; Kirilovsky, D.; Spezia, R.; Robert, B. *Biochim. Biophys. Acta, Bioenerg.* **2015**, *1847* (10), 1044–1054.
- Polivka, T.; Chabera, P.; Kerfeld, C. A. *Biochim. Biophys. Acta, Bioenerg.* **2013**, *1827* (3), 248–254.
- Niedzwiedzki, D. M.; Liu, H. J.; Blankenship, R. E. *J. Phys. Chem. B* **2014**, *118* (23), 6141–6149.
- Gwizdala, M.; Wilson, A.; Kirilovsky, D. *Plant Cell* **2011**, *23* (7), 2631–2643.
- Leverenz, R. L.; Jallet, D.; Li, M. D.; Mathies, R. A.; Kirilovsky, D.; Kerfeld, C. A. *Plant Cell* **2014**, *26* (1), 426–437.
- Zhang, H.; Liu, H.; Niedzwiedzki, D. M.; Prado, M.; Jiang, J.; Gross, M. L.; Blankenship, R. E. *Biochemistry* **2014**, *53* (1), 13–19.
- Teo, Y. N.; Kool, E. T. *Chem. Rev.* **2012**, *112* (7), 4221–4245.
- Albinsson, B.; Hannestad, J. K.; Borjesson, K. *Coord. Chem. Rev.* **2012**, *256* (21–22), 2399–2413.
- Spillmann, C. M.; Medintz, I. L. *J. Photochem. Photobiol., C* **2015**, *23*, 1–24.
- Dutta, P. K.; Levenberg, S.; Loskutov, A.; Jun, D.; Saer, R.; Beatty, J. T.; Lin, S.; Liu, Y.; Woodbury, N. W.; Yan, H. *J. Am. Chem. Soc.* **2014**, *136* (47), 16618–16625.
- Boulay, C.; Wilson, A.; D'Haene, S.; Kirilovsky, D. *Proc. Natl. Acad. Sci. U. S. A.* **2010**, *107* (25), 11620–11625.
- Sutter, M.; Wilson, A.; Leverenz, R. L.; Lopez-Igual, R.; Thurotte, A.; Salmeen, A. E.; Kirilovsky, D.; Kerfeld, C. A. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, *110* (24), 10022–10027.